GATE DA 2027

Probability & Statistics

Topic-by-topic notes with formulas and the stories of Ramu and Somu

Meet the two friends.
Ramu is curious and asks "but why?" about everything. **Somu** is his friend who is good at maths and explains with small everyday examples. In every chapter Ramu asks a question, Somu answers with a story, and then we write the clean formulas.

How each chapter is organised: story → idea in plain words → formulas → worked example → common traps.

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1 Counting (Permutations and Combinations)

Ramu & Somu: Outfits, cricket team and laddus

Ramu has **3 shirts** and **2 pants**. "How many different outfits can I wear?" Somu says: for each shirt you can choose any pant, so $3 \times 2 = 6$ outfits. This is the **product rule**: if one job can be done in m ways *and then* another in n ways, together they can be done in $m \times n$ ways. Next, Ramu has **5 friends** and wants to pick **2** of them for a doubles match. Does the order matter? Picking (Somu, Ravi) is the same team as (Ravi, Somu), so order does not matter. The count is $\binom{5}{2} = 10$.
But if Ramu must choose a **captain and a vice-captain** from those 5, order matters (captain Somu, vice Ravi is different from captain Ravi, vice Somu). Count: $5 \times 4 = 20 = P(5, 2)$.

Idea in plain words

Before counting, ask two questions: (1) **Does order matter?** (2) **Can things repeat?** Those two answers decide the formula.

1.1 Basic rules

Formulas

Sum rule (either this OR that, disjoint): $m + n$

Product rule (this THEN that): $m \times n$

$n! = n(n-1)\cdots 2 \cdot 1, \qquad 0! = 1$

1.2 Permutations and combinations

Formulas

Situation	Count
Arrange r of n distinct items (order matters, no repeat)	$P(n, r) = \dfrac{n!}{(n-r)!}$
Choose r of n (order does not matter)	$\binom{n}{r} = \dfrac{n!}{r!\,(n-r)!}$
r-length strings from n symbols, repetition allowed	n^r
Choose r of n with repetition, order irrelevant	$\binom{n+r-1}{r}$
Arrange n items where $n_1, \ldots, n_k$ are identical	$\dfrac{n!}{n_1!\, n_2! \cdots n_k!}$
Circular arrangement of n distinct items	$(n-1)!$

Ramu & Somu: Why divide for identical letters?

Ramu asks: "How many different words can I make from the letters of `MISSISSIPPI`?" Somu says: pretend all letters are different, giving 11! arrangements. But the four S's are identical, so swapping them does not create a new word. Those 4! swaps all look the same, so divide by 4!. Do the same for the four I's (4!) and two P's (2!):

$$\frac{11!}{4!\,4!\,2!} = \frac{39916800}{1152} = 34650.$$

1.3 Useful identities

Formulas

Symmetry, Pascal: $\binom{n}{r} = \binom{n}{n-r}$, $\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n, \qquad \sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0$$

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}, \qquad k\binom{n}{k} = n\binom{n-1}{k-1}$$

$$\sum_{k} \binom{m}{k}\binom{n}{r-k} = \binom{m+n}{r} \quad \text{(Vandermonde)}$$

1.4 Stars and bars

Ramu & Somu: Sharing laddus

Somu has **10 identical laddus** and wants to give them to **3 friends**: Ramu, Ravi and Raju. A friend may get zero. How many ways? Draw the 10 laddus as stars and use 2 bars to cut them into 3 groups, e.g. $\star\star\star|\star\star\star\star|\star\star\star$ means (3,4,3). We are choosing positions of 2 bars among $10 + 2 = 12$ slots:

$$\binom{12}{2} = 66.$$

If **everyone must get at least one**, first give each friend 1 laddu (3 used), then share the remaining 7 freely: $\binom{9}{2} = 36$.

Formulas

Solutions of $x_1 + x_2 + \cdots + x_k = n$:

non-negative integers: $\binom{n+k-1}{k-1}$ positive integers: $\binom{n-1}{k-1}$.

If $x_i \geq a_i$ for some variables, substitute $y_i = x_i - a_i$ and use the non-negative formula.

1.5 Inclusion–exclusion, derangements, pigeonhole

Formulas

$$|A \cup B| = |A| + |B| - |A \cap B|$$
$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$
$$D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!} \approx \frac{n!}{e} \qquad (D_1 = 0,\ D_2 = 1,\ D_3 = 2,\ D_4 = 9,\ D_5 = 44)$$

Pigeonhole: put N objects into k boxes $\Rightarrow$ some box has at least $\lceil N/k \rceil$ objects.

Ramu & Somu: Secret Santa gone wrong

Ramu, Somu and two more friends each bring a gift and put them in a bag; each randomly picks one. "What is the chance nobody gets his own gift?" Somu counts **derangements**: $D_4 = 9$ out of $4! = 24$ arrangements, so the probability is $9/24 = 0.375$. For large n this settles near $1/e \approx 0.368$, which is surprising: it hardly depends on n.

Common GATE traps

(1) Mixing up "arrange" (P) and "choose" (C). (2) Forgetting to divide for identical objects. (3) In probability, use the *same* model (ordered or unordered) in numerator and denominator.

2 Probability Axioms, Sample Space and Events

Ramu & Somu: Ramu tosses a coin

Ramu tosses a coin. Somu says: list everything that can happen: $S = \{H, T\}$. This list is the **sample space**. "Getting a head" is an **event**, a subset of S. If the coin is fair, each outcome is equally likely so $\mathbb{P}(H) = 1/2$. In general, for equally likely outcomes

$$\mathbb{P}(A) = \frac{|A|}{|S|} = \frac{\text{favourable outcomes}}{\text{all outcomes}}.$$

2.1 Kolmogorov's three axioms

Axioms

(1) $\mathbb{P}(A) \geq 0$ for every event A

(2) $\mathbb{P}(S) = 1$

(3) If $A_1, A_2, \ldots$ are mutually exclusive: $\mathbb{P}\Big(\bigcup_i A_i\Big) = \sum_i \mathbb{P}(A_i)$

2.2 What follows from the axioms

Formulas

$$\mathbb{P}(\varnothing) = 0, \quad \mathbb{P}(A^c) = 1 - \mathbb{P}(A), \quad 0 \leq \mathbb{P}(A) \leq 1$$

$$A \subseteq B \Rightarrow \mathbb{P}(A) \leq \mathbb{P}(B)$$

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

$$\mathbb{P}(A \cup B \cup C) = \sum \mathbb{P}(A_i) - \sum \mathbb{P}(A_i \cap A_j) + \mathbb{P}(A \cap B \cap C)$$

$$\mathbb{P}(A \setminus B) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$$

Union bound: $\mathbb{P}\Big(\bigcup_i A_i\Big) \leq \sum_i \mathbb{P}(A_i)$

De Morgan: $(A \cup B)^c = A^c \cap B^c$

$(A \cap B)^c = A^c \cup B^c$

Ramu & Somu: Cricket and football in the class

In Ramu's class, 60% like cricket, 50% like football and 80% like at least one of them. Ramu asks: "What percentage like both?" Somu: if you simply add $60 + 50 = 110\%$ you counted the "both" students twice, so subtract the overlap:

$$0.8 = 0.6 + 0.5 - \mathbb{P}(C \cap F) \;\Rightarrow\; \mathbb{P}(C \cap F) = 0.3.$$

Since $\mathbb{P}(C)\mathbb{P}(F) = 0.6 \times 0.5 = 0.3$ equals $\mathbb{P}(C \cap F)$, liking cricket and liking football are *independent* in this class.

2.3 Mutually exclusive versus independent

Idea in plain words

Mutually exclusive means $A \cap B = \varnothing$: they cannot happen together. **Independent** means that knowing one tells you nothing about the other: $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. If both have positive probability they can *never* be both. Example: on one coin toss "head" and "tail" are exclusive, but if Ramu knows it is a head he is 100% sure it is not a tail, so they are strongly dependent.

Worked example

Two fair dice are rolled. $\mathbb{P}(\text{sum} = 7) = 6/36 = 1/6$. $\mathbb{P}(\text{sum} \geq 10) = (3 + 2 + 1)/36 = 1/6$. $\mathbb{P}(\text{at least one six}) = 1 - (5/6)^2 = 11/36$.

3 Conditional Probability, Independence and Bayes' Theorem

Ramu & Somu: The bag of balls

Ramu's bag has 3 red and 2 blue balls. He draws one ball and **does not put it back**, then draws another. Somu: "What is the chance both are red?" The first is red with probability 3/5. After removing one red, 2 red remain among 4 balls, so the second is red with probability 2/4. Multiply:

$$\mathbb{P}(R_1 \cap R_2) = \mathbb{P}(R_1)\mathbb{P}(R_2 \mid R_1) = \frac{3}{5} \cdot \frac{2}{4} = \frac{3}{10}.$$

The **conditional probability** $\mathbb{P}(R_2 \mid R_1)$ means: the chance of R_2 *given that* R_1 has already happened. The information "first was red" changed the bag, so the probability changed.

3.1 Definitions

Formulas

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \qquad \mathbb{P}(B) > 0$$

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A)$$

$$\text{Chain rule: } \mathbb{P}(A_1 \cap \cdots \cap A_n) = \mathbb{P}(A_1)\mathbb{P}(A_2 \mid A_1)\mathbb{P}(A_3 \mid A_1A_2)\cdots\mathbb{P}(A_n \mid A_1 \cdots A_{n-1})$$

$$\mathbb{P}(A^c \mid B) = 1 - \mathbb{P}(A \mid B)$$

3.2 Independence

Formulas

$$A \perp\!\!\!\perp B \iff \mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) \iff \mathbb{P}(A \mid B) = \mathbb{P}(A)$$

$$A \perp\!\!\!\perp B \Rightarrow A \perp\!\!\!\perp B^c,\ A^c \perp\!\!\!\perp B,\ A^c \perp\!\!\!\perp B^c$$

Events A, B, C are **mutually independent** only if *all* of these hold:

$$\mathbb{P}(AB) = \mathbb{P}(A)\mathbb{P}(B),\ \mathbb{P}(AC) = \mathbb{P}(A)\mathbb{P}(C),\ \mathbb{P}(BC) = \mathbb{P}(B)\mathbb{P}(C),\ \mathbb{P}(ABC) = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C).$$

Ramu & Somu: Pairwise is not enough

Ramu and Somu each toss a fair coin. Let A: Ramu gets H; B: Somu gets H; C: both coins show the same face. Check: $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = 1/2$ and each pair has probability $1/4$, so every pair is independent. But $\mathbb{P}(A \cap B \cap C) = 1/4 \neq 1/8$. So the three are *pairwise* independent but not *mutually* independent: knowing A and B completely decides C.

3.3 Total probability and Bayes

Idea in plain words

Sometimes it is easier to split a problem into cases. If $B_1, \dots, B_k$ are disjoint cases that together cover everything (a **partition**), the probability of A is the weighted average of how likely A is in each case. Bayes' theorem then *reverses* the conditioning: from "how likely is the evidence given the cause" to "how likely is the cause given the evidence".

Formulas

$$\text{Total probability: } \mathbb{P}(A) = \sum_{i=1}^{k} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)$$

$$\text{Bayes: } \mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}$$

$$\text{Odds form: posterior odds} = \text{prior odds} \times \text{likelihood ratio}$$

Names: $\mathbb{P}(B_j)$ = **prior**, $\mathbb{P}(A \mid B_j)$ = **likelihood**, $\mathbb{P}(B_j \mid A)$ = **posterior**. Naive Bayes and Bayesian networks (AI section) are built on this.

Ramu & Somu: The scary medical test

A rare disease affects 1% of people. A test says "positive" for 99% of the sick people (sensitivity) and also wrongly for 5% of healthy people. Ramu tests positive and panics. Somu calms him by counting a group of 10,000 people:

- Sick: 100 people $\Rightarrow$ 99 test positive.
- Healthy: 9900 people $\Rightarrow$ $0.05 \times 9900 = 495$ test positive.

Among all $99 + 495 = 594$ positives, only 99 are really sick: $99/594 = 1/6 \approx 0.167$. Formally,

$$\mathbb{P}(D \mid +) = \frac{0.99 \times 0.01}{0.99 \times 0.01 + 0.05 \times 0.99} = \frac{0.0099}{0.0594} = \frac{1}{6}.$$

The test is "99% accurate" for sick people, yet a positive result means only about a 17% chance of disease, because healthy people are so much more numerous.

Two machines

Machine M_1 makes 60% of items with 2% defective; M_2 makes 40% with 5% defective. Then $\mathbb{P}(\text{def}) = 0.6(0.02) + 0.4(0.05) = 0.012 + 0.020 = 0.032$, and $\mathbb{P}(M_1 \mid \text{def}) = 0.012/0.032 = 0.375$.

Common GATE traps

(1) $\mathbb{P}(A \mid B) \neq \mathbb{P}(B \mid A)$ in general (the "prosecutor's fallacy"). (2) The Bayes denominator must be summed over a full partition. (3) Drawing without replacement makes draws dependent.

4 Random Variables, Expectation and Variance

Ramu & Somu: A game with Somu

Somu offers Ramu a game: roll a fair die and receive Rs. equal to the number shown. "What do I get on average?" Somu: the number rolled is a **random variable** X (a number decided by chance). Each value $1, \dots, 6$ has probability $1/6$, so the long-run average is

$$\mathbb{E}[X] = 1 \cdot \tfrac{1}{6} + 2 \cdot \tfrac{1}{6} + \cdots + 6 \cdot \tfrac{1}{6} = 3.5.$$

You never actually get 3.5 on one roll, but over many rolls your average approaches it. This is the **expectation**.

4.1 Types and functions

Formulas

$$\text{Discrete: } p(x) = \mathbb{P}(X = x), \quad p(x) \geq 0, \quad \sum_x p(x) = 1$$

$$\text{Continuous: } f(x) \geq 0, \quad \int_{-\infty}^{\infty} f(x)\,dx = 1, \quad \mathbb{P}(a \leq X \leq b) = \int_a^b f(x)\,dx$$

$$\text{CDF: } F(x) = \mathbb{P}(X \leq x);\ \text{non-decreasing, right-continuous, } F(-\infty) = 0,\ F(\infty) = 1$$

$$\mathbb{P}(a < X \leq b) = F(b) - F(a), \qquad f(x) = F'(x) \text{ (continuous)}$$

For a continuous variable a single point has probability zero, $\mathbb{P}(X = x) = 0$; probability lives in intervals (areas under the curve).

4.2 Expectation and variance

Formulas

$$\mathbb{E}[X] = \sum_x x\,p(x) \quad \text{or} \quad \int x f(x)\,dx$$

$$\mathbb{E}[g(X)] = \sum_x g(x)p(x) \quad \text{or} \quad \int g(x)f(x)\,dx \qquad \text{(LOTUS)}$$

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b, \qquad \mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y] \quad \text{(always, even if dependent)}$$

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] \quad \text{if } X \perp\!\!\!\perp Y \quad \text{(converse false)}$$

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2, \qquad \sigma = \sqrt{\mathrm{Var}(X)}$$

$$\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X), \qquad X \perp\!\!\!\perp Y \Rightarrow \mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$$

$$\text{Tail formula: } X \geq 0 \Rightarrow \mathbb{E}[X] = \int_0^{\infty} \mathbb{P}(X > x)\,dx \quad \Big(\text{integers: } \sum_{k \geq 1} \mathbb{P}(X \geq k)\Big)$$

$$\text{MGF: } M_X(t) = \mathbb{E}[e^{tX}], \quad \mathbb{E}[X^k] = M_X^{(k)}(0), \quad X \perp\!\!\!\perp Y \Rightarrow M_{X+Y} = M_X M_Y$$

Idea in plain words

Expectation is the *centre of mass*; variance measures the *spread*. Adding a constant moves the centre but does not change the spread. Multiplying by a stretches the spread by a, so the variance is multiplied by a^2.

Ramu & Somu: Is Somu's gamble fair?

Somu proposes: with probability 0.3 Ramu wins Rs. 10, otherwise (probability 0.7) he loses Rs. 3. Then

$$\mathbb{E}[X] = 10(0.3) + (-3)(0.7) = 3 - 2.1 = 0.9, \qquad \mathbb{E}[X^2] = 100(0.3) + 9(0.7) = 36.3,$$

$$\mathrm{Var}(X) = 36.3 - 0.9^2 = 35.49.$$

On average Ramu gains Rs. 0.90 per game, but the large variance says that single games are very swingy.

Ramu & Somu: The indicator trick: hats

n people throw hats in a pile and each picks one at random. Ramu asks for the expected number of people who get their own hat. Somu writes $X = I_1 + \cdots + I_n$ where $I_i = 1$ if person i gets his own hat. Each $\mathbb{P}(I_i = 1) = 1/n$, so by linearity

$$\mathbb{E}[X] = n \cdot \frac{1}{n} = 1 \quad \text{for every } n,$$

even though the I_i are *not* independent. Linearity of expectation never needs independence.

Worked example

$f(x) = 2x$ on $[0,1]$. $\mathbb{E}[X] = \int_0^1 2x^2 dx = \frac{2}{3}$, $\mathbb{E}[X^2] = \int_0^1 2x^3 dx = \frac{1}{2}$, so $\mathrm{Var}(X) = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}$. Median m: $F(m) = m^2 = \frac{1}{2} \Rightarrow m = 1/\sqrt{2}$.

5 Common Discrete Distributions

Ramu & Somu: Penalty kicks

Ramu scores a penalty with probability $p = 0.7$. Each kick is a **Bernoulli trial**: success (1) with probability p, failure (0) with probability $q = 1 - p$. Somu then asks different questions about the same situation, and each question gives a named distribution:

- "Out of 5 kicks, how many goals?" $\to$ **Binomial**$(5, 0.7)$.
- "How many kicks until his first goal?" $\to$ **Geometric**(0.7).
- "How many kicks until his 3rd goal?" $\to$ **Negative binomial**.

5.1 Summary table

Distribution	PMF	Mean	Var	Notes
Uniform $\{1..n\}$	$\frac{1}{n}$	$\frac{n+1}{2}$	$\frac{n^2-1}{12}$	fair die: 3.5, $\frac{35}{12}$
Bernoulli(p)	$p^x(1-p)^{1-x}$	p	pq	one trial
Binomial(n,p)	$\binom{n}{k}p^k q^{n-k}$	np	npq	# successes in n trials; mode $\lfloor (n+1)p \rfloor$
Geometric(p)	$q^{k-1}p,\ k \geq 1$	$\frac{1}{p}$	$\frac{q}{p^2}$	trials to 1st success; memoryless
Neg. binomial(r,p)	$\binom{k-1}{r-1}p^r q^{k-r}$	$\frac{r}{p}$	$\frac{rq}{p^2}$	trials to r-th success
Poisson(λ)	$e^{-\lambda}\lambda^k/k!$	λ	λ	rare events per interval
Hypergeo.(N,K,n)	$\binom{K}{k}\binom{N-K}{n-k}/\binom{N}{n}$	$\frac{nK}{N}$	$n\frac{K}{N}\left(1-\frac{K}{N}\right)\frac{N-n}{N-1}$	no replacement

5.2 Binomial

Ramu & Somu: Five kicks

Ramu takes 5 penalties, $p = 0.7$. What is the probability of exactly 3 goals? Choose which 3 of the 5 kicks are goals: $\binom{5}{3} = 10$ ways. Each such pattern has probability $0.7^3 \cdot 0.3^2$. So

$$\mathbb{P}(X = 3) = \binom{5}{3}(0.7)^3(0.3)^2 = 10 \times 0.343 \times 0.09 = 0.3087.$$

Mean $= np = 3.5$ goals, variance $= npq = 5(0.7)(0.3) = 1.05$.

5.3 Geometric and memorylessness

Ramu & Somu: Waiting for the first head

Somu keeps tossing a fair coin until the first head. Expected number of tosses: $1/p = 2$. Ramu says: "I have already got 5 tails, so a head is surely due!" Somu laughs: the coin has no memory.

$$\mathbb{P}(X > m + n \mid X > m) = \mathbb{P}(X > n) = q^n.$$

The past failures do not change the future waiting time. Be careful: some books define geometric as the number of *failures* before the first success (mean q/p); read the question.

5.4 Poisson

Ramu & Somu: Messages per minute

Somu sends Ramu on average $\lambda = 2$ messages per minute, at random moments. The number in one minute is Poisson(2): $\mathbb{P}(0) = e^{-2} \approx 0.135$, $\mathbb{P}(2) = e^{-2} \cdot \frac{2^2}{2!} \approx 0.271$. For a Poisson variable mean and variance are both λ.

Relationships to remember

$\mathrm{Bin}(n, p) =$ sum of n independent Bernoulli(p)

$X \sim \mathrm{Bin}(n_1, p),\ Y \sim \mathrm{Bin}(n_2, p)$ independent $\Rightarrow X + Y \sim \mathrm{Bin}(n_1 + n_2, p)$

$X \sim \mathrm{Poi}(\lambda_1),\ Y \sim \mathrm{Poi}(\lambda_2)$ independent $\Rightarrow X + Y \sim \mathrm{Poi}(\lambda_1 + \lambda_2)$

Poisson approximation: n large, p small $\Rightarrow \mathrm{Bin}(n, p) \approx \mathrm{Poi}(\lambda = np)$

Poisson ratio: $\dfrac{\mathbb{P}(k+1)}{\mathbb{P}(k)} = \dfrac{\lambda}{k+1}$ (useful for the mode)

5.5 Hypergeometric

Ramu & Somu: Cards without replacement

A box holds 10 cards, 4 red and 6 black. Ramu draws 3 cards *without replacement*. Probability of exactly 2 red:

$$\frac{\binom{4}{2}\binom{6}{1}}{\binom{10}{3}} = \frac{6 \times 6}{120} = 0.3.$$

If he drew *with* replacement it would be Binomial; without replacement it is hypergeometric and its variance carries the factor $\frac{N-n}{N-1}$.

Common GATE traps

Poisson has mean = variance; if a problem gives different values it cannot be Poisson. Check the geometric convention. For hypergeometric remember the finite-population correction.

6 Common Continuous Distributions

Ramu & Somu: Waiting for a bus

Ramu reaches a bus stop at a random time; the bus comes at some moment in the next 20 minutes, with no moment more likely than another: waiting time $\sim \text{Uniform}(0, 20)$. Somu: $\mathbb{P}(\text{wait} > 15) = \frac{5}{20} = 0.25$, mean wait $= 10$ min, variance $= \frac{20^2}{12} \approx 33.3$.

6.1 Summary

Distribution	PDF / CDF	Mean	Var
Uniform(a, b)	$f = \frac{1}{b-a}$ on $[a, b]$; $F = \frac{x-a}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential(λ)	$f = \lambda e^{-\lambda x}$, $x \geq 0$; $F = 1 - e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Normal(μ, σ^2)	$f = \frac{1}{\sigma\sqrt{2\pi}} \exp\Big(-\frac{(x-\mu)^2}{2\sigma^2}\Big)$	μ	σ^2
Gamma(α, λ)	$f \propto x^{\alpha-1} e^{-\lambda x}$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$

6.2 Exponential distribution

Ramu & Somu: How long does Somu's phone battery last?

Suppose the lifetime is exponential with mean 100 hours, so $\lambda = 1/100$. Ramu: "Chance it lasts more than 50 hours?" $\mathbb{P}(X > 50) = e^{-50/100} = e^{-0.5} \approx 0.607$. Ramu: "It has already lasted 80 hours, is it about to die?" Somu: for an exponential, the remaining life has the *same* distribution no matter how old it is. That is the **memoryless** property, the continuous cousin of the geometric.

Formulas

$$\mathbb{P}(X > x) = e^{-\lambda x}, \qquad \mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t)$$

$$\min(X_1, \ldots, X_k) \sim \text{Exp}(\lambda_1 + \cdots + \lambda_k) \quad \text{for independent } X_i \sim \text{Exp}(\lambda_i)$$

Poisson process of rate λ : counts in time $t \sim \text{Poi}(\lambda t)$, gaps between events $\sim \text{Exp}(\lambda)$

6.3 Normal distribution

Ramu & Somu: Exam marks

Marks in the class follow $N(60, 10^2)$. Ramu scored 80. Somu **standardises**: $z = \frac{80-60}{10} = 2$, i.e. Ramu is 2 standard deviations above average. $\mathbb{P}(X > 80) = \mathbb{P}(Z > 2) = 1 - \Phi(2) = 1 - 0.9772 = 0.0228$, so about the top 2.3%. The cutoff for the top 5% is $60 + 1.645 \times 10 = 76.45$.

Formulas

$$Z = \frac{X-\mu}{\sigma} \sim N(0,1), \qquad \mathbb{P}(X \le x) = \Phi\Big(\frac{x-\mu}{\sigma}\Big)$$

$$\Phi(-z) = 1 - \Phi(z), \qquad \mathbb{P}(|Z| \le z) = 2\Phi(z) - 1$$

$$aX + b \sim N(a\mu + b,\ a^2\sigma^2)$$

$$X \perp\!\!\!\perp Y:\ X + Y \sim N(\mu_1 + \mu_2,\ \sigma_1^2 + \sigma_2^2) \quad \text{(variances add, never SDs)}$$

Empirical rule: $\pm 1\sigma \approx 68.3\%,\ \pm 2\sigma \approx 95.4\%,\ \pm 3\sigma \approx 99.7\%$

$$\Phi(1.28) = 0.90,\ \Phi(1.645) = 0.95,\ \Phi(1.96) = 0.975,\ \Phi(2.33) = 0.99,\ \Phi(2.576) = 0.995$$

6.4 Distributions built from the normal

Formulas

$$\chi^2_k = Z_1^2 + \cdots + Z_k^2: \quad \text{mean } k,\ \text{variance } 2k$$

$$t_\nu = \frac{Z}{\sqrt{\chi^2_\nu/\nu}}: \quad \text{mean } 0\ (\nu > 1),\ \text{variance } \frac{\nu}{\nu-2}\ (\nu > 2),\ \ t_\nu \to N(0,1) \text{ as } \nu \to \infty$$

$$F = \frac{\chi^2_m/m}{\chi^2_n/n}$$

Normal sample: $\dfrac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}, \qquad \dfrac{\bar{X}-\mu}{S/\sqrt{n}} \sim t_{n-1}$

7 Joint Distributions, Covariance and Correlation

Ramu & Somu: Ramu's and Somu's study hours

Let $X = 1$ if Ramu studies tonight and $Y = 1$ if Somu studies. They live together, so they influence each other. The joint probabilities are:

	$Y=0$	$Y=1$	$\mathbb{P}(X=x)$
$X=0$	0.3	0.2	0.5
$X=1$	0.1	0.4	0.5
$\mathbb{P}(Y=y)$	0.4	0.6	1

Row and column totals are the **marginals**. Are X, Y independent? $\mathbb{P}(X=0, Y=0) = 0.3$ but $0.5 \times 0.4 = 0.2$, so *no*. Now compute: $\mathbb{E}[X] = 0.5$, $\mathbb{E}[Y] = 0.6$, $\mathbb{E}[XY] = \mathbb{P}(X=1, Y=1) = 0.4$, so

$$\operatorname{Cov}(X,Y) = 0.4 - 0.5 \times 0.6 = 0.1 > 0,$$

positive: when Ramu studies, Somu tends to as well. With $\operatorname{Var} X = 0.25$, $\operatorname{Var} Y = 0.24$:

$$\rho = \frac{0.1}{\sqrt{0.25 \times 0.24}} \approx 0.408.$$

Formulas

Marginal: $p_X(x) = \sum_y p(x,y)$ or $\int f(x,y)\,dy$, Conditional: $p(y \mid x) = \dfrac{p(x,y)}{p_X(x)}$

$X \perp\!\!\!\perp Y \iff p(x,y) = p_X(x)p_Y(y)$ for all x, y

$\operatorname{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$, $\operatorname{Cov}(X,X) = \operatorname{Var}(X)$

$\operatorname{Cov}(aX+b, cY+d) = ac\,\operatorname{Cov}(X,Y)$

$\operatorname{Var}(X \pm Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) \pm 2\operatorname{Cov}(X,Y)$

$\rho_{XY} = \dfrac{\operatorname{Cov}(X,Y)}{\sigma_X \sigma_Y} \in [-1,1]$, $|\rho| = 1 \iff Y = aX + b$

Total expectation: $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]]$

Total variance: $\operatorname{Var}(X) = \mathbb{E}[\operatorname{Var}(X \mid Y)] + \operatorname{Var}(\mathbb{E}[X \mid Y])$

Idea in plain words

Independent ⇒ uncorrelated (covariance 0), but **not the other way round**. Example: let X be symmetric about 0 and $Y = X^2$. Then $\operatorname{Cov}(X,Y) = \mathbb{E}[X^3] - \mathbb{E}[X]\mathbb{E}[X^2] = 0$, yet Y is completely determined by X. Exception: for *jointly normal* variables, uncorrelated $\iff$ independent.

Worked example

For n i.i.d. variables with mean μ and variance σ^2: sum has mean $n\mu$, variance $n\sigma^2$; sample mean $\bar{X}$ has mean μ and variance σ^2/n. If $\operatorname{Var} X = 4$, $\operatorname{Var} Y = 9$, $\rho = 0.5$ then $\operatorname{Cov} = 0.5 \cdot 2 \cdot 3 = 3$ and $\operatorname{Var}(X+Y) = 4 + 9 + 2(3) = 19$.

8 Inequalities, Law of Large Numbers and Central Limit Theorem

Ramu & Somu: How much can we say without knowing the distribution?

Ramu: "The average salary in a company is Rs. 50,000. How many people can earn Rs. 5,00,000 or more?" Somu: at most 10%. If more than 10% earned that much, the average would already exceed Rs. 50,000. This is **Markov's inequality**: using only the mean we can bound a tail.

8.1 Bounds

Formulas

$$\text{Markov } (X \geq 0,\ a > 0): \quad \mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

$$\text{Chebyshev:} \quad \mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}, \quad \text{i.e.} \quad \mathbb{P}(|X - \mu| \geq c) \leq \frac{\sigma^2}{c^2}$$

$$\text{Jensen } (f \text{ convex}): \quad f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)] \quad (\text{so } \mathbb{E}[X^2] \geq (\mathbb{E}X)^2)$$

Chebyshev example: mean 50, SD 5. Then $\mathbb{P}(40 < X < 60) \geq 1 - \frac{1}{2^2} = 0.75$.

8.2 Law of Large Numbers

Ramu & Somu: Many tosses

Ramu tosses a coin 10 times and gets 7 heads: 70%. Somu says: toss it 10,000 times and the proportion will be very close to 50%. The **law of large numbers**: for i.i.d. X_i with mean μ, the sample mean $\bar{X}_n \to \mu$ as $n \to \infty$. It does *not* say tails "catch up"; it says the early deviation gets diluted.

8.3 Central Limit Theorem

Ramu & Somu: Why bell curves are everywhere

Somu: take any random thing (even a very skewed one), average many independent copies, and the average looks like a bell curve. That is the **CLT**.

Formulas

For i.i.d. $X_1, \ldots, X_n$ with mean μ and variance σ^2 (any shape), for large n (rule of thumb $n \geq 30$):

$$\bar{X} \approx N\Big(\mu,\ \frac{\sigma^2}{n}\Big), \qquad S_n = \sum X_i \approx N(n\mu,\ n\sigma^2)$$

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \ \to\ N(0,1), \qquad \text{standard error } \ \text{SE} = \frac{\sigma}{\sqrt{n}}$$

Normal approximation to Binomial ($np \geq 5$, $nq \geq 5$) with continuity correction:

$$\mathbb{P}(X \leq k) \approx \Phi\Big(\frac{k + 0.5 - np}{\sqrt{npq}}\Big), \qquad \mathbb{P}(X \geq k) \approx 1 - \Phi\Big(\frac{k - 0.5 - np}{\sqrt{npq}}\Big).$$

100 coin tosses

$X \sim \text{Bin}(100, 0.5)$, $np = 50$, $\sqrt{npq} = 5$. $\mathbb{P}(X \geq 60) \approx 1 - \Phi(\frac{59.5-50}{5}) = 1 - \Phi(1.9) = 1 - 0.9713 \approx 0.029$.

Sample mean of marks

Marks have $\mu = 20$, $\sigma = 12$; a sample of $n = 36$ has SE $= 12/6 = 2$. $\mathbb{P}(\bar{X} > 23) = \mathbb{P}(Z > 1.5) = 1 - 0.9332 = 0.0668$.

Common GATE traps

Markov and Chebyshev give *bounds*, not exact values. The CLT is about means and sums, not that raw data become normal. Use $\sigma/\sqrt{n}$, not σ, for the spread of the sample mean.

9 Sampling, Estimation and Confidence Intervals

Ramu & Somu: Estimating the class average height

Ramu wants the average height of all students in college (the **population mean** μ), but he cannot measure everyone. Somu measures 25 random students and gets a sample mean $\bar{x} = 165$ cm. This number is an **estimate** of μ. Ramu: "But if you pick another 25 students you get a different number!" Exactly: that is why we attach a **margin of error**, giving an interval instead of a single number.

9.1 Point estimation

Formulas

$$\bar{x} = \frac{1}{n}\sum x_i, \qquad s^2 = \frac{1}{n-1}\sum (x_i - \bar{x})^2 \quad \text{(divide by } n-1 \Rightarrow \text{unbiased for } \sigma^2)$$

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta, \qquad \text{MSE} = \text{Var}(\hat{\theta}) + \text{Bias}^2$$

$$\text{MLE: maximise } L(\theta) = \prod f(x_i;\theta) \text{ (or } \log L)$$

$$\text{Bernoulli: } \hat{p} = \bar{x};\ \text{Poisson: } \hat{\lambda} = \bar{x};\ \text{Exponential: } \hat{\lambda} = 1/\bar{x};$$

$$\text{Normal: } \hat{\mu} = \bar{x},\ \hat{\sigma}^2 = \tfrac{1}{n}\sum (x_i - \bar{x})^2 \text{ (biased)}$$

9.2 Confidence intervals

Confidence interval of level $1-\alpha$

Parameter / setting	Interval
Mean, σ known (or large n)	$\bar{x} \pm z_{\alpha/2}\dfrac{\sigma}{\sqrt{n}}$
Mean, σ unknown, normal data	$\bar{x} \pm t_{n-1,\alpha/2}\dfrac{s}{\sqrt{n}}$
Proportion (large n)	$\hat{p} \pm z_{\alpha/2}\sqrt{\dfrac{\hat{p}(1-\hat{p})}{n}}$
Variance (normal data)	$\left[\dfrac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}},\ \dfrac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}}\right]$
Difference of means, σ's known	$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2}\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}$

Common $z_{\alpha/2}$: 90% $\to$ 1.645, 95% $\to$ 1.96, 99% $\to$ 2.576.

Sample size for margin of error E: $n = \left(\dfrac{z_{\alpha/2}\,\sigma}{E}\right)^2$ (round *up*).

Ramu & Somu: What does "95% confident" mean?

Somu builds a 95% interval for the average height: (160.1, 169.9). Ramu: "So there is a 95% chance the true mean is in this interval?" Somu: careful. The true mean is a fixed number; it is either inside or not. The 95% describes the *method*: if we repeated the whole sampling process many times, about 95 out of 100 such intervals would capture the true mean. More data $\Rightarrow$

narrower interval; higher confidence ⇒ wider interval.

Worked example

$\bar{x} = 54$, $\sigma = 10$, $n = 25$: SE $= 2$. 95% CI $= 54 \pm 1.96(2) = (50.08,\ 57.92)$.
Sample size for $E = 1$ at 95%: $n = (1.96 \times 10/1)^2 = 384.16 \to \mathbf{385}$.
Large sample, $\bar{x} = 80$, $s = 15$, $n = 36$: $80 \pm 1.96(15/6) = 80 \pm 4.9 = (75.1,\ 84.9)$.

10 Hypothesis Testing (z-test, t-test, Chi-square)

Ramu & Somu: The pizza delivery claim

A shop says its pizza arrives in **30 minutes on average**. Ramu doubts it, so Somu records 25 deliveries and finds an average of 34 minutes. Is this just bad luck, or is the claim false? Somu sets up a *courtroom*:

- H_0 (the accused is innocent): $\mu = 30$. We keep it unless the evidence is strong.
- H_1 (the charge): $\mu \neq 30$ (or > 30 for a one-sided charge).
- The evidence is the sample mean. If it would be *very unlikely* under H_0, we reject H_0.

Type I error: convicting an innocent person (reject a true H_0), probability α. **Type II error:** freeing a guilty person (keep a false H_0), probability β. **Power** $= 1 - \beta$.

10.1 Procedure

1. State H_0 (contains "=") and H_1 (one- or two-sided).
2. Choose the significance level α (commonly 0.05).
3. Compute the test statistic assuming H_0 is true.
4. Compare with the critical value, or compute the p-value. Reject H_0 if $|\text{stat}| >$ critical, equivalently $p \leq \alpha$.

Formulas

$\alpha = \mathbb{P}(\text{Type I}), \quad \beta = \mathbb{P}(\text{Type II}), \quad \text{Power} = 1 - \beta \ \ (\uparrow \text{ with larger } n, \text{ larger effect, larger } \alpha)$

p-value $= \mathbb{P}(\text{statistic at least as extreme as observed} \mid H_0 \text{ true})$

10.2 z-tests (when σ is known, or n large)

Formulas

$$\text{One mean: } z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

$$\text{Two means: } z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$$

$$\text{One proportion: } z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$$

Critical values: two-sided $\alpha = 0.05 \to \pm 1.96$, $\alpha = 0.01 \to \pm 2.576$; one-sided 1.645 (5%), 2.33 (1%).

Pizza, with σ known

$H_0 : \mu = 30$, $\sigma = 10$, $n = 25$, $\bar{x} = 34$. $z = \frac{4}{10/5} = 2 > 1.96$, so **reject** H_0 at 5%. $p \approx 2(1 - 0.9772) = 0.0456$. (The matching 95% CI $34 \pm 3.92 = (30.08, 37.92)$ excludes 30, which agrees with the test.)

10.3 t-tests (when σ is unknown, small n, normal data)

Formulas

$$\text{One sample: } t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad \text{df} = n - 1$$

$$\text{Paired: } t = \frac{\bar{d}}{s_d/\sqrt{n}}, \quad \text{df} = n - 1 \quad (d = \text{differences})$$

$$\text{Two sample (equal variances): } s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{1/n_1 + 1/n_2}}, \quad \text{df} = n_1 + n_2 - 2$$

$$\text{Welch (unequal variances): } t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

Two-sided 5% critical values $t_{0.025,\text{df}}$: df $= 4 \to 2.776$, $9 \to 2.262$, $15 \to 2.131$, $19 \to 2.093$, $29 \to 2.045$, $\infty \to 1.96$.

Small sample

$n = 16$, $\bar{x} = 52$, $s = 8$, $\mu_0 = 50$: $t = \frac{2}{8/4} = 1.0$, df $= 15$, critical 2.131. Not extreme enough, so **fail to reject** H_0.
Another: $n = 9$, $\bar{x} = 105$, $s = 12$, $\mu_0 = 100$: $t = \frac{5}{12/3} = 1.25 < 2.306$ (df $= 8$): fail to reject.

10.4 Chi-square tests

Ramu & Somu: Is Somu's die fair?

Ramu thinks Somu's die is loaded. They roll 60 times and get counts $5, 8, 9, 8, 10, 20$ for faces $1, \ldots, 6$. A fair die should give 10 of each. Somu measures the total "distance" between observed (O) and expected (E):

$$\chi^2 = \sum \frac{(O - E)^2}{E} = \frac{25 + 4 + 1 + 4 + 0 + 100}{10} = 13.4.$$

Degrees of freedom $= 6 - 1 = 5$; the 5% critical value is 11.07. Since $13.4 > 11.07$, **reject** fairness: face 6 appears suspiciously often.

Formulas

$$\text{Goodness of fit: } \chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

$$\text{df} = (\#\text{categories}) - 1 - (\#\text{estimated parameters})$$

$$\text{Independence } (r \times c \text{ table}): E_{ij} = \frac{(\text{row total})(\text{column total})}{\text{grand total}}$$

$$\text{df} = (r - 1)(c - 1)$$

$$\text{Variance of a normal population: } \chi^2 = \frac{(n - 1)s^2}{\sigma_0^2}, \quad \text{df} = n - 1$$

Goodness-of-fit and independence tests are right-tailed; each E_i should be at least about 5. Handy: for df $= 1$, critical value (5%) $= 3.84$.

Independence, 2×2

Observed $\begin{pmatrix} 30 & 20 \\ 20 & 30 \end{pmatrix}$, all margins 50, grand total 100, so every $E = 25$. $\chi^2 = 4 \times \frac{(5)^2}{25} = 4.0 > 3.84$ with df $= 1$: reject independence.

Common GATE traps

(1) "Fail to reject H_0" does *not* prove H_0 true. (2) A two-sided test at level α agrees with the $(1-\alpha)$ CI. (3) Use t when σ is unknown and n is small, z when σ is given. (4) Paired data $\Rightarrow$ paired t. (5) Lowering α reduces Type I error but raises Type II error (for fixed n).

11 One-page Formula Sheet

Topic	Remember
Counting	$\binom{n}{r} = \frac{n!}{r!(n-r)!}$; stars and bars $\binom{n+k-1}{k-1}$; $D_n \approx n!/e$; identical items $\frac{n!}{n_1!\cdots n_k!}$
Probability	$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$; $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$
Conditional / Bayes	$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(AB)}{\mathbb{P}(B)}$; $\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A\|B_j)\mathbb{P}(B_j)}{\sum_i \mathbb{P}(A\|B_i)\mathbb{P}(B_i)}$
$\mathbb{E}$, Var	$\operatorname{Var} X = \mathbb{E}X^2 - (\mathbb{E}X)^2$; $\operatorname{Var}(aX+b) = a^2 \operatorname{Var} X$; $\mathbb{E}(X+Y) = \mathbb{E}X + \mathbb{E}Y$
Binomial / Poisson / Geometric	np, npq / λ, λ / $1/p$, q/p^2
Uniform / Exponential / Normal	$\frac{a+b}{2}$, $\frac{(b-a)^2}{12}$ / $\frac{1}{\lambda}$, $\frac{1}{\lambda^2}$ / μ, σ^2
Covariance	$\mathbb{E}[XY] - \mathbb{E}X\,\mathbb{E}Y$; $\operatorname{Var}(X+Y) = \operatorname{Var} X + \operatorname{Var} Y + 2\operatorname{Cov}(X,Y)$
Inequalities	Markov $\frac{\mathbb{E}X}{a}$; Chebyshev $\frac{1}{k^2}$
CLT	$\bar{X} \approx N(\mu, \sigma^2/n)$; Bin $\approx N(np, npq)$ with ± 0.5
Confidence interval	$\bar{x} \pm z\frac{\sigma}{\sqrt{n}}$ or $\bar{x} \pm t\frac{s}{\sqrt{n}}$
Tests	$z = \frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}}$; t with df $= n-1$; $\chi^2 = \sum \frac{(O-E)^2}{E}$

12 Practice Questions (with answers)

Try each on your own first.

1. In how many ways can Ramu, Somu and 3 other friends sit around a circular table? **Ans:** $4! = 24$.
2. $\mathbb{P}(A) = 0.6$, $\mathbb{P}(B) = 0.5$, $\mathbb{P}(A \cup B) = 0.8$. Are A, B independent? **Ans:** $\mathbb{P}(A \cap B) = 0.3 = 0.6 \times 0.5$: yes.
3. Machine M_1 (60% output, 2% defective), M_2 (40%, 5% defective). A defective item is found. $\mathbb{P}(M_2)$? **Ans:** $0.020/0.032 = 0.625$.
4. $X \sim \text{Bin}(10, 0.3)$. Find mean and variance. **Ans:** 3 and 2.1.
5. Calls arrive as Poisson, mean 4 per hour. $\mathbb{P}$(no call in an hour)? **Ans:** $e^{-4} \approx 0.0183$.
6. $X \sim$ Exp with mean 10. $\mathbb{P}(X > 5)$? **Ans:** $e^{-0.5} \approx 0.607$.
7. $X \sim N(60, 10^2)$. $\mathbb{P}(X < 75)$? **Ans:** $\Phi(1.5) = 0.9332$.
8. $\operatorname{Var} X = 4$, $\operatorname{Var} Y = 9$, $\rho = 0.5$. Find $\operatorname{Var}(X+Y)$. **Ans:** $4 + 9 + 2(3) = 19$.
9. $\mu = 20$, $\sigma = 12$, $n = 36$. $\mathbb{P}(\bar{X} > 23)$? **Ans:** $\mathbb{P}(Z > 1.5) \approx 0.0668$.
10. $\bar{x} = 80$, $s = 15$, $n = 36$. Approximate 95% CI? **Ans:** $(75.1,\ 84.9)$.
11. $n = 9$, $\bar{x} = 105$, $s = 12$, $H_0 : \mu = 100$ (two-sided, 5%). Decision? **Ans:** $t = 1.25 < 2.306$: fail to reject.
12. Observed $\begin{pmatrix} 30 & 20 \\ 20 & 30 \end{pmatrix}$. Test independence at 5%. **Ans:** $\chi^2 = 4.0 > 3.84$: reject.

How to study this section: learn the chapters in order, derive each formula once yourself, then solve 15–20 GATE-style questions per chapter and keep an error log. The highest-yield areas are

Bayes, expectation/variance of named distributions, covariance, normal/exponential calculations, CLT approximations and choosing between z, t and χ^2 tests. Always check table values against the statistical tables you will actually use.